CS 2300 HW 9

1a) Let us assume that there exists a set S such that:

S = {x ∈ Z | x is divisible by 4} = {....,-4,0,4,8,12,....}

Let us enumerate the elements of the set S as follows:

0,4,-4,8,-8,12,-12,16,-16,....

Let us assume that there exists another bijective function f such that:

f: N → S
f(n) = { 4*n/2, if n is even }
f(n) = (-4 * (n+1)/2, if n is odd}

For various values of n, we can define f(n) as follows:

f(0) = 4 * 0/2 = 0
f(1) = -4 * ((1+1)/2) = -4
f(2) = 4 * (2/2) = 4
f(3) = -4 * (3+1)/2 = -4 * (4/2) = -4 * 2 = -8
f(4) = 4 * (4/2) = 8
f(5) = -4 * (5+1)/2 = -4 * (6/2) = -4 * 3 = -12
….(etc)

Therefore, we can say that this function assigns every natural number to a unique element in a set of integers that are divisible by 4 and every such integer appears at least one time in the function.

b) By looking at the question, we can say that $A\ X\ Z^{+}$, where A is the Cartesian product of a finite set and a countably infinite set:

$$A\ X\ Z^{+} = \{(a, z) \mid a \in \{0, 1, 2, 3\},\ z \in Z^{+}\}$$

We can enumerate it as follows:

(0,1),(1,1),(2,1),(3,1),(0,2),(1,2)(2,2),(3,2),(0,3),(1,3),(2,3),(3,3)....

Let us define a bijective function g such that:

$$g(n) = N -> Z^{+}$$
g(n) = [n mod 4, (n/4) + 1]

Now, for various values n, g can be represented as following:

g(0) = (0,1)
g(1) = (1,1)
g(2) = (2,1)
g(3) =(3,1)
g(4) =(0,1)
g(5) = (1,2)
g(6) = (2,2)
….(etc)

Therefore, we can say that g consists of every possible pair in $A\ X\ Z^{+}$

2a) **Finite**

Let us assume that A = R and B = R - {1,2,3}, where R represents all real numbers.

Then, if we do A - B, then we get:

A - B = {1,2,3}

We get {1,2,3} as our answer because A consists of all real numbers and B consists of all real numbers except for {1,2,3} so when we A-B, it gives the elements that are present in A but not in B.

b) **Countably Infinite**

Let us assume that A = R and B = R - Q, where R represents all real numbers and Q represents all rational numbers.

Then, if we do A - B, then we get:

A - B = Q

We get Q as our answer because A consists of all real numbers and B consists of all irrational numbers so when we do A-B, it gives the elements that are present in A but not in B and A consists of all rational numbers and B doesn’t so that’s why we get Q as an answer.

c) **Uncountably Infinite**

Let us assume that A = R and B = [0,1], where R represents all real numbers.

Then, if we do A - B, then we get:

$$A - B = (-\infty,0) \cup (1,\infty)$$

We get $(-\infty,0) \cup (1,\infty)$ as our answer because A consists of all real numbers and B of all the real numbers that lie in between 0 and 1,so when we do A-B, it gives the elements that are present in A but not in B.

3) According to the question, we can say that:

$$\sum = \{0,1,2,3,4,5,6,7,8,9,/ \}$$

Let us assume that:

$$\sum^* = \{x: \text{x is a finite string with symbols from } \sum \}.$$

Let us consider a set $\sum^n$, where $n \in N$ and N represents all natural numbers:

$$\sum^n = \{\text{all strings of length n}\}$$

Since $| \sum | = 11$, we can say that $\sum^n$ has 11^n elements which are considered as a finite amount.

Since $\sum^n$ is finite, there may be many countable n's, so we can say that $\sum^*$ is a countable union of finite sets. Thus, we can say that $\sum^*$ is countable.

We can represent every positive rational in the form of p/q, where p,q $\in N^+$

Since we aren't talking about strings over here we can write it as:

"Digits of p"/"Digits of q" $\in \sum^*$.

Hence, we have an injection over here.

Also, since $Q^+ \rightarrow \sum^*$ and p/q can be represented as (decimal digits of p)/(decimal digits of q),we can say that Q^+ is countable because $\sum^*$ is countable, so any of its subsets are going to be countable.

4) Let us assume that there exists a set F, which is a set of all functions from N to N such that:

$$F = \{f:N \rightarrow N\}$$

Contradiction rule

Let us assume that F is countable, then we can list its functions as follows:

$$f_0, f_1, f_2, f_3, \dots.$$

Over here we can say that, for each f_k: N→N, every function in F appears somewhere in this list.

Let us define a new function g such that:

g: N → N
g(n) = f_n(n) +1

Let us say that there exists any index k that belongs to a natural number such that at any point n=k it can be represented as follows:

g(k) = f_k(k) + 1

Since fk(k)+1 is not equal to f_k(k),so we can say that g(k) is not equal to f_k(k), and so we can say that g and fk are not the same functions and as this holds true for every k, the function g cannot coincide with any fk in our list.

Hence, we can say that our assumption that every function in F was on our list is wrong. Thus, we can say that no enumeration of F can exist by natural numbers, hence F is uncountable.

5) Let us say that there exists a set S such that it consists of all real numbers whose digits are all 3's.

Let us say that there exists some element x such that x ∈ S and it has the form:

x = 33…3.33……3

Let us consider two variables m and n such that m represents the number of 3's that come in the first part of the number and n represents the number of 3's that come in the second part of the number.

In x, we can say that m must be greater than or equal to 1 and n must be greater than or equal to 0(if we consider n to be 0, then we can say that there is no fractional part in the number like m=4 and n=0, gives us 3333)

Since we know that N x N is countable by antidiagonal enumeration, we can list all the pairs (m,n) in the order of m+n as follows:

m + n = 0: (1,0)
m + n =1: (1,1),(2,0)
m + n = 2: (1,2),(2,2),(3,1),(4,0) and so on…

Hence, we can say that whenever the enumeration reaches (m,n), we can say that the output can be a real number.

Hence, $x_{(m,n)}$ = 3333…3333..3333.

Therefore, we can say that every element of x appears exactly one time in this list,m so we can say that s is countable.

b) Let us consider a set T which only contains real numbers whose decimal digits are only 3 or 4.

Let us assume that T can be listed as follows:

$$x^{(1)}, x^{(2)}, x^{(3)},...$$

Where each $x^{(k)}$ = 0.$d_{k,1}d_{k,2}d_{k,3}$… has $d_{k,j}$ ∈ {3,4}

Let us define a new decimal y such that:

y= 0 $y_1y_2y_3$… by y_n = 3, if $d_{n,n}$ = 4 and y_n = 4 if $d_{n,n}$ = 3

Over here, in simple words we can say that, $y_n$basically flips the nth digit of the nth listed number, so if each y_n ∈ {3,4}, then we can say that y also belongs to T.

Therefore, we can say that by construction y differs from $x^{(n)}$ in the nth decimal place, so y is not equal to x^n for every n.

Therefore, we can say that y is a member of T that is not in our supposed complete list, which proves to be a contradiction to our assumption, so that’s why we can say that T is uncountable.

6) Over here, we can say that the proof is attempting to show that N is uncountable by writing each natural number a_i as an infinite decimal sequence, which includes a tail of zeroes as well, then we can define it as follows:

b_j = (d_{jj}+ 1) mod 10

Over here we can observe a contradiction, that is when are claiming b to be a new natural number.

Now, let us see what has gone wrong, by the definition we can say that a valid decimal representation of a natural number must only have finite non zero digits, that is, it must eventually have all zeroes, but the constructed diagonal b has b_j =1 for all large values of j, so it never becomes 0 and does not represent any natural number and thus there must be no contradiction.